Difference in complexity of language used by writers from different socioeconomic backgrounds

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github.com/tjfeen/special-barnacle

Abstract

Writers from different socioeconomic backgrounds have a different target audience and a different writing style. Although a lot of work has been done before on the complexity of works targeted to audiences from various socioeconomic backgrounds, less research has been done in relation to the background of writers. Therefore, the present paper focuses on the text complexity of literary works by writers from different socioeconomic backgrounds.

1 Introduction

The history of writing literature has a lot of interesting aspects, especially when studying variations in language. One could research how the language used in literature targeted towards people from higher socioeconomic classes differs from that of literature written for people from a less prosperous background. Because a lot of research has already been done on that specific topic, this report focuses on the less-described flip side of the coin: the difference in language used by *writers* from different socioeconomic backgrounds. The hypothesis is as follows: literary works from upper-class writers contain more complex language, compared to works of writers from a less fortunate background. The independent variable is the writer's background, and the dependent variable is the complexity of the language used by said writer.

2 Related Work

Not a lot of work has been published on this specific topic, making it hard to find related peer-reviewed studies, although some less-closely related papers were found.

Michelle Yu has conducted research at the Southern New Hampshire University on how women, from the nineteenth century all the way through to the modern era, were able to overcome the diffculties raised by society. Professional writing was most often done by only men, and women had the sole task of birthing and raising children, and they weren't allowed to have an opinion of their own. Yu (2021)

Although the paper does not focus on text complexity at all, it is interesting as to how the real-life diffculties faced by a (female) writer influence in their literary works.

Another interesting study Lawton (2006) focuses not necessarily on language used by professional writers, but that of boys from both working- and middle-class. Twenty boys, ten from both backgrounds, were asked to write essays about four topics, 30 minutes for each topic. The first noticable difference between the essays produced by the boys is the length: the working-class boys produce significantly shorter essays than the middle-class boys. In addition to that, the middle-class boys used more passive verbs (although the difference is small). The middle-class boys also used less common words compared to the working-class boys.

Although text length is not relevant for the present study, the present hypothesis is supported by this paper; well-educated writers, from a more fortunate socioeconomic background, *on average*, write more complex texts than writers from the lower, working-class.

An earlier study Bernstein (1960) finds that members from the lower working-class use language that is more descriptive and concrete, whereas middle-class members use more abstract language.

3 Data

The dataset used contains works from American writer Mark Twain, representing the lower- and middle-class writers, and Irish writer Oscar Wilde, representing the upper-class writers. For each writer, the top three most downloaded books, as of 10-01-2025, from Project Gutenberg were used, respectively *Adventures of Huckleberry Finn*, *The Adventures of Tom Sawyer*, *The Prince and the Pauper* and *The Picture of Dorian Gray*, *The Importance of Being Earnest: A Trivial Comedy for Serious People*, *De Profundis*.

Pre-processing For each book, the plain-text variant was downloaded. The metadata of the book, Project Gutenberg's usage license as well as anything leading up to the first chapter (including the table of contents) were manually removed, such that only the words written by the author remained. Additionally, any characters not matching [A-z.,!;:?-] were removed.

For books written in dialogue, the following rules were added as well:

- Ignore words placed in square brackets, such as *[Enter Jack.]* and *[Opens case and examines it.]*;
- Ignore lines containing only one uppercase word, e.g. *ALGERNON*.

For each book a complexity score is calculated as follows:

- *s*: the average sentence length in words;
- *w*: the average length of all unique words.
- *p*: the average number of comma's, colons and semi-colons in a sentence;

To calculate the overall complexity score *c*, the following formula is used: $s + w + 2p$. This yields complexity scores that are typically between 20 and 40, where a higher number indicates a more complex text.

4 Results

Discussion It would be reasonable to think, and so did we, that writers from a better socioeconomic background would have had better education, therefore writing works with more complex language, compared to writers from a less fortunate background.

	label	s	w	p	c
1	wilde-earnest	11.5	6.8	0.7	19.8
2	wilde-picture-dorian	14.1	6.9	1.0	22.9
3	twain-sawyer	18.4	7.0	1.5	28.3
4	wilde-profundis	24.2	6.5	1.7	34.1
5	twain-huckleberry	23.6	6.5	2.1	34.3
6	twain-prince-pauper	26.8	6.9	2.5	38.9

Table 1: s, w, p and c-scores for each of the six works, sorted by overall complexity.

However, looking at the results in Table 1, we learn that upper-class works (represented by Wilde), have a lower average complexity score of *25.6*, whereas the lower- and middle class works (represented by Twain) have an average complexity score of *33.9*, meaning the hypothesis is not supported by the results from the research process.

The way of measuring text complexity might not be the most accurate, and could be improved by comparing the words to a set of words from standard, every-day English to measure the number of words in common between the sets after lemmatization. The number of passive verbs could also be another factor to text complexity. Perhaps language models could be used as well to study the text complexity.

In addition to that, using three works by a writer to represent all works by all the writers from the same background does not really yield clear results, and therefore this could also be improved by using more works, perhaps grabbing random pages from each work to limit the size of the dataset.

5 Conclusion

This study aimed at discovering the difference in text complexity by authors from various socioeconomic backgrounds, found that authors from less fortunate background write more complex texts, and better-educated writers write texts that are less complex, meaning the present hypothesis is *not* supported.

The method of measuring text complexity could be improved by using language models to actually interpret the text, as the present paper looked more at sentence length, word length and the use of punctuation.

References

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